HUMAN-CENTRIC ARTIFICIAL INTELLIGENCE

Project 4: Preference Elicitation

A reproducible movie-preference study comparing repeated pairwise choices with ten-item rankings. This report documents Tasks 1-3 and the implemented participant interface.

METHODS AND PLANNED USER STUDY

Version 1.2 | 18 August 2026 | Study not conducted

SECTION 01

Executive summary

The project asks how quickly a new-user movie recommender can learn a latent preference vector from limited interaction. It compares a low-complexity pairwise interface with a more expressive ten-movie ranking interface. Both methods use the same movie metadata, feature map, utility function, regularization and held-out evaluation set.

Deliverable status

The Django prototype implements the landing page, downloadable report, consent, both elicitation interfaces, short workload questionnaires, held-out validation, server-side logging and model fitting. No participants were recruited and no empirical results are claimed.

Research question

After equal exposure to 20 movies, does a Plackett-Luce model trained on two complete ten-item rankings predict new pairwise movie choices more accurately than a Bradley-Terry model trained on ten pairwise choices, and what additional cognitive burden does ranking impose?

Design at a glance

Element	Pairwise condition	Ranking condition
Participant action	Choose one of two movies	Order ten movies from best to worst
Movie exposure	10 pairs = 20 unique movies	2 lists = 20 unique movies
Preference model	Bradley-Terry	Plackett-Luce
Primary comparison	Mean log loss on the same 20 unseen validation pairs	Same validation responses
Experience measures	Time, mental demand, confidence, ease	Time, mental demand, confidence, ease

Human-centric commitments

- No IMDb score, vote count or social-media popularity is shown or learned as a taste signal.
- No name, email, free-text demographic detail, IP address or third-party analytics is required.
- Ranking supports drag-and-drop and keyboard move controls; visible focus and high contrast are retained.
- The study order is counterbalanced and validation responses never update either model.

SECTION 02

Movie feature representation - Task 1

Source and preprocessing

The implementation uses the IMDb 5000 Movie Dataset file linked directly from the assignment brief. The bundled file contains 5,043 rows and 28 metadata fields but no individual user ratings. The documented filters retain 4,789 distinct, usable films. Titles are cleaned of trailing non-breaking spaces; rows without a title, genre, director, plausible year or plausible duration are removed; duplicate title-year pairs keep the most widely voted record. Sampling is seeded and without replacement inside a participant session.

Feature vector x

Feature block	Encoding	Why it belongs
Genres	Multi-hot	Directly represents stable thematic preferences.
Release decade	One-hot	Captures preference for eras without assuming a strictly linear year effect.
Duration	Dataset z-score	Represents pacing and time commitment on a comparable numeric scale.
Language	Top 8 + OTHER	Captures viewing-language preference while controlling dimensionality.
Country	Top 10 + OTHER	Adds broad production context without sparse country indicators.
Content rating	Frequent levels + OTHER	Represents content constraints relevant to watch choice.
Plot keywords	TF-IDF, then 16-component SVD	Adds themes beyond genre while keeping the low-data model compact.

Deliberate exclusions

- IMDb score, number of votes, gross, budget and Facebook counters are popularity or exposure proxies, not personal taste.
- Raw director and actor one-hot features create thousands of sparse dimensions that cannot be estimated reliably from 10 choices or 2 rankings; the names remain visible on cards as decision context.
- Movie title is an identifier, not a transferable content feature. It is never encoded into x.

Shared representation

Both conditions use the exact same cached float64 feature matrix. Feature extraction is fit once on metadata only; participant choices never affect preprocessing. The utility has no intercept because a constant cancels in all utility differences.

SECTION 03

Pairwise preference model

For movie i with feature vector x_i and participant preference vector w, utility is linear. The Bradley-Terry model turns the utility difference between two films into a choice probability.

```
u_i = w^T x_i
```

```
P(i preferred to j | w) = sigmoid(w^T (x_i - x_j))
```

For each observed winner i and loser j, d = x_i - x_j. The implementation minimizes the L2-regularized negative log likelihood below. Regularization is essential because the number of observed choices is much smaller than the feature dimension.

```
J_BT(w) = sum_t log(1 + exp(-d_t^T w)) + (lambda / 2) ||w||_2^2
```

```
grad J_BT(w) = -sum_t d_t sigmoid(-d_t^T w) + lambda w
```

Numerical implementation

- SciPy L-BFGS-B with analytic gradient, w initialized at zero and lambda fixed at 1.0 before data collection.
- numpy.logaddexp and scipy.special.expit avoid overflow for large positive or negative utilities.
- The participant never sees model probabilities, which prevents prediction feedback from changing later choices.
- Unit tests require probability symmetry, 0.5 for equal features, finite optimization and recovery of a known synthetic direction.

Interpretation

A positive weight indicates that the participant tends to select films containing that feature, conditional on all other encoded attributes. Magnitude reflects model scale and regularization; it should not be interpreted as a causal effect. Correlated genres and keyword topics may share weight.

Cold-start scope

The fitted w belongs to one pseudonymous participant and is estimated only from that participant's study interactions. The model does not use collaborative ratings or transfer preferences between users.

SECTION 04

Ranking extension - Task 2

A ten-film ordering is not treated as 45 independent binary observations. Those implied pairs come from one correlated action. Instead, the implementation uses the Plackett-Luce model, a stagewise extension of Bradley-Terry. At each stage the participant selects the best remaining item.

```
P(i_1 > ... > i_m | w) = product_(r=1)^(m-1) exp(u_(i_r)) / sum_(s=r)^m exp(u_(i_s))
```

The first factor is the probability that i_1 is selected from all m films; the second selects i_2 from the remaining $m-1$; the process continues until one film remains. When $m = 2$, the single factor reduces exactly to the Bradley-Terry probability. This nesting gives a direct theoretical justification for comparing the two interfaces.

```
J_PL(w) = sum_q sum_(r=1)^(m_q-1) [logsumexp_(s=r..m_q)(u_(i_s)) - u_(i_r)] + (lambda /
2)||w||_2^2
```

Information and dependence

One ten-item ranking contains 45 ordered item pairs, but the Plackett-Luce likelihood contains nine sequential choice stages. The report records both counts and never claims 45 statistically independent responses. Ranking can encode more preference structure per movie exposure, while requiring more working memory and interaction effort. That trade-off motivates the user study.

Validation tests

- A two-item Plackett-Luce fit must match Bradley-Terry loss and parameters within numerical tolerance.
- Rankings must contain every presented movie exactly once; malformed JSON, duplicates and foreign IDs are rejected server-side.
- The objective uses logsumexp and must remain finite on extreme synthetic utilities.
- Repeated synthetic orderings must produce utilities in the expected order.

SECTION 05

User study design - Task 3

Hypotheses

ID	Prespecified statement	Outcome
H1 primary	After equal exposure to 20 movies, ranking produces lower held-out pairwise log loss than pairwise elicitation.	Participant-level mean negative log loss on 20 common validation pairs.
H2	Ranking tasks take longer and are more mentally demanding.	Block time and 1-7 mental-demand response.
H3	Ranking produces higher confidence that responses express the participant's taste.	1-7 confidence response.
Exploratory	Interface preference is associated with workload and predictive advantage.	Preferred method and optional comment.

Within-subject counterbalanced design

Every participant completes both methods and therefore serves as their own control, which is important because movie preferences vary strongly between people. Server-side allocation alternates pairwise-first and ranking-first sequences to maintain AB/BA balance. In a real run, allocation should use concealed permuted blocks generated before recruitment.

Sequence	Block 1	Block 2	Validation
AB	10 pairwise choices	2 ten-film rankings	20 unseen pairwise choices
BA	2 ten-film rankings	10 pairwise choices	20 unseen pairwise choices

The two elicitation pools and the validation pool are disjoint within each participant. Each elicitation condition exposes exactly 20 unique movies. Films are sampled uniformly without replacement from cleaned records with a title, genre, director, plausible year and plausible duration, and at least 1,000 historical votes; vote count is used only as an eligibility filter to reduce unfamiliar or extremely obscure stimuli, and is neither displayed nor encoded. A sensitivity run should remove this filter.

SECTION 06

Participants, recruitment and procedure

Recruitment and sample size

The target population is adults who can read English and make movie choices on a laptop, desktop or large tablet. Recruitment would use university mailing lists and campus notices, with no course credit controlled by the research team and with compensation stated before consent. Participants must be at least 18, provide informed consent, and not have completed the study previously.

Planned sample

For a paired standardized effect d_z = 0.40, two-sided alpha = 0.05 and power = 0.80, the planned minimum is 52 complete participants. Recruit 62 because ceiling(52 / 0.85) = 62, preserving the target after approximately 15 percent attrition or technical exclusion. First run a 10-12 person feasibility pilot; pilot data are not pooled with confirmation data.

Participant journey

Step	Stage	Action
1	Landing	Read neutral purpose, expected duration and privacy summary; download the full report if desired.
2	Consent	Confirm age, voluntary participation and local storage of pseudonymous responses/timings.
3	First block	Complete the assigned elicitation method, followed by mental demand, confidence and ease ratings.
4	Second block	Complete the other method with a disjoint movie pool, followed by the same ratings.
5	Validation	Make 20 pairwise choices on unseen films; these choices update neither model.
6	Final feedback	Choose a preferred interface, report movie-watching frequency and optionally explain the choice.
7	Debrief	Receive a random study code and explanation of why model scores are hidden from participants.

Exclusion rules fixed before analysis

- No valid consent or age confirmation; duplicate participation; incomplete elicitation block; or fewer than 16 of 20 validation choices.
- Corrupt allocation, missing movie IDs or impossible timestamps are technical exclusions.
- Very fast choices and repeated selection of one screen side are flagged for sensitivity analysis, not automatic exclusion.

SECTION 07

Measures and prespecified analysis

Outcome construction

After both blocks, two separate models are fit with the same feature matrix and lambda = 1.0. For every validation pair, each model outputs the probability assigned to the participant's chosen movie. Primary log loss is the negative mean log probability across the 20 trials. Lower values indicate better calibrated prediction.

Family	Measure	Role
Predictive	Validation log loss	Primary
Predictive	Accuracy and Brier score	Secondary
Efficiency	Total response time per block and per movie exposure	Secondary
Experience	Mental demand, confidence, ease of use (1-7)	Secondary
Preference	Preferred interface and optional explanation	Exploratory

Primary analysis

For each participant compute Delta = NLL_ranking - NLL_pairwise. Test the mean paired difference with a two-sided alpha of .05 and report the 95 percent confidence interval and paired effect size d_z. The directional hypothesis predicts Delta < 0, but a two-sided test avoids understating an unexpected reverse effect. A paired permutation test is the prespecified robustness analysis.

Secondary analysis

- Compare accuracy with a trial-level logistic mixed model containing method, period and participant random intercept.
- Compare response time on the log scale and Likert responses with paired ordinal or Wilcoxon analyses.
- Apply Holm correction across the three confirmatory secondary hypotheses.
- Report complete-case results plus a sensitivity analysis including participants with at least 80 percent validation completion.
- Do not tune features, lambda or optimizer settings on validation answers.

No results section

This is a planned protocol and executable interface. Because the study has not been conducted, the report specifies estimands and analysis requirements without presenting outcome estimates.

SECTION 08

Implementation, logging and safeguards

Server-side workflow

Django stores a random UUID in the browser session and keeps the authoritative study stage in SQLite. A seeded movie plan is created once at consent. Every POST is CSRF-protected, IDs are checked against the current server-side stimulus set, database writes are atomic and unique constraints prevent duplicate trials.

Table	Stored fields	Purpose
StudySession	UUID, order, stage, seed, movie plan, timestamps	Counterbalancing, resume and reproducibility
PairwiseChoice	kind, trial, left/right/chosen IDs, duration	Elicitation and held-out validation
RankingResponse	presented IDs, permutation, duration	Plackett-Luce fitting and integrity
BlockFeedback	method, demand, confidence, ease	Human experience outcomes
FinalFeedback	preferred method, movie frequency, optional comment	Interpretation and exploratory analysis

Privacy and ethics

- The prototype collects no direct identifiers and does not intentionally log IP address, browser fingerprint or third-party analytics.
- Choice and timing data remain pseudonymous rather than truly anonymous; real deployment needs a named data controller, legal basis, retention period, withdrawal process and access controls.
- Research consent and the GDPR legal basis are separate decisions. A real study requires supervisor review and the relevant TUHH ethics/data-protection process before recruitment.
- Participants may stop at any time. A production study should offer a deletion request using the random completion code until a stated deadline.

Interface acceptance criteria

- Landing page exposes both required actions: download the report and start the study.
- Pairwise trials show exactly two equal-weight cards; ranking trials contain exactly ten unique movies.
- Ranking supports mouse drag interaction and keyboard up/down buttons; server validation remains authoritative.
- Visible progress, clear instructions, responsive layout, reduced-motion support and WCAG-oriented focus/contrast are included.

SECTION 09

Limitations, reproducibility and references

Threats to validity

- Metadata cards cannot reproduce the rich decision context of trailers, posters or prior viewing; familiarity may dominate some choices.
- The historical-vote eligibility filter improves recognizability but underrepresents obscure and non-mainstream films.
- Linear utility and the Plackett-Luce independence-of-irrelevant-alternatives assumption may miss context effects and non-compensatory taste.
- Two ten-item rankings versus ten binary choices intentionally differ in information density; conclusions apply to equal movie exposure, not equal interaction time.
- A convenience university sample would limit generalization by age, culture and movie-consumption patterns.

Reproducibility checklist

- Dataset path and required schema are validated; the assignment-linked CSV is bundled with provenance notes.
- Movie sampling seed, method order and all presented IDs are stored per participant.
- Feature order is deterministic; keyword SVD uses random_state = 0; model lambda and optimizer are fixed.
- Tests cover dataset integrity, seeded disjoint sampling, model identities, malformed ranking input, CSRF and the complete study flow.
- Run with: python manage.py migrate; python manage.py test project4; python manage.py runserver.

References

[1] Human-Centric Artificial Intelligence. Project 4: Preference elicitation. Course assignment brief, pp. 1-2.

[2] Bradley, R. A., and Terry, M. E. (1952). Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons. Biometrika 39(3/4), 324-345. doi:10.1093/biomet/39.3-4.324.

[3] Plackett, R. L. (1975). The Analysis of Permutations. Applied Statistics 24(2), 193-202. doi:10.2307/2346567.

[4] Luce, R. D. (1959). Individual Choice Behavior: A Theoretical Analysis. New York: Wiley.

[5] IMDb 5000 Movie Dataset file linked by the assignment. GitHub dataset page.

[6] European Union (2016). General Data Protection Regulation, Regulation (EU) 2016/679. Official text.

Implementation note

The participant-facing interface is written in English to match the course brief. This report documents what the prototype does and clearly separates implemented behavior from recommended procedures for a future real deployment.